

Tennis with Thompson

PROJECT DESCRIPTION & PLAN

USA Men's Tennis: 2026 Match-by-Match Analysis

A reproducible Python and Jupyter Notebook project for analyzing American men's performance at ATP Masters 1000 events and Grand Slams

Project type	Sports analytics, automated data pipeline, and interactive reporting
Analysis period	2026 season to date
Population	U.S. male players appearing in selected ATP Masters 1000 and Grand Slam main draws
Primary tools	Python, Jupyter Notebook, pandas, NumPy, Altair, and TennisMyLife CSV data
Primary output	A player-centric analytical CSV, summary tables, match drill-downs, and interactive charts

Executive Summary

This project evaluates the match-by-match performance of American male tennis players during the 2026 ATP season at the sport's largest tournaments: Grand Slams and Masters 1000 events. It combines completed-season records with ongoing tournament results, validates and deduplicates the source data, and restructures each match into an American-player perspective. The result is a reusable analytical dataset that supports both player-level comparisons and detailed review of individual matches.

It measures serve effectiveness, break-point performance, ranking upsets, surface results, tournament progression, deciding-set and tiebreak outcomes, comeback wins, and actual performance relative to ranking-based expectations. Interactive Altair visualizations allow the user to explore results without relying on static charts.

1. Project Purpose and Objectives

The purpose of the project is to create a reliable and repeatable framework for monitoring how U.S. men's tennis players perform against elite competition throughout the 2026 season.

- Consolidate current and completed 2026 match data into one validated source.
- Identify every qualifying match involving at least one American male player.
- Standardize winner-loser records into a consistent player-versus-opponent format.
- Measure results, serving performance, break-point execution, ranking upsets, and clutch performance.
- Compare actual outcomes with transparent ranking-based expectations.
- Provide interactive summaries that can be refreshed manually as the season progresses.

2. Scope

Scope area	Definition
Included	2026 ATP men's main-draw matches at Grand Slams and Masters 1000 tournaments; matches involving one or two U.S. players; completed and currently ongoing events.
Excluded	ATP 250 and 500 events, Challenger events, qualifying draws, women's matches, doubles, and seasons before 2026.
Country rule	A match qualifies when winner_ioc or loser_ioc equals USA.
Event rule	Events must match the notebook's approved event list and carry the Grand Slam or Masters level in the 2026 source files.
Refresh rule	The notebook refreshes on demand when REFRESH_DATA is set to True and all cells are rerun.

3. Data Source

The project uses CSV files published by the TennisMyLife Tennis Match Database. The notebook requests both the completed 2026 ATP file and the ongoing-tournaments file. Successful downloads are cached locally, allowing the notebook to fall back to the most recent saved copy if a later request fails.

Source page: <https://stats.tennismylife.org/tennis-match-database>

Key source fields include tournament identifiers, event name, date, level, surface, round, player IDs, nationality codes, rankings, score, match duration, aces, double faults, serve-point totals, first- and second-serve results, and break points faced and saved.

4. How the Project Works

1. **Configure the run:** The notebook creates raw and processed data folders, defines the source URLs, establishes the approved event list, and controls whether fresh data should be requested.
2. **Download and cache data:** Completed and ongoing CSV files are downloaded. When a request fails, the notebook uses an existing cached file and reports that fallback to the user.
3. **Validate the schema:** Required columns are checked before analysis. Missing fields trigger an error instead of allowing incomplete calculations to proceed silently.
4. **Clean and deduplicate:** Completed and live files are combined. Matches are assigned stable keys, and overlapping records are resolved by preferring the more current and complete row.
5. **Filter the project population:** The pipeline retains 2026 Grand Slam and Masters 1000 matches from the approved tournament list, then identifies records involving at least one U.S. player.
6. **Reshape to player perspective:** Winner and loser columns are normalized so every analytical row uses the same player and opponent fields. An all-American match creates two rows, one from each player's perspective.
7. **Engineer analytical metrics:** The notebook calculates rankings, serve rates, break-point measures, upset indicators, set-level outcomes, tiebreak performance, and ranking-based expectations.
8. **Export and visualize:** The player-centric dataset is saved as a processed CSV and used to build summary tables, player drill-downs, and interactive Altair charts.
9. **Run final quality checks:** Assertions verify uniqueness, valid outcomes, U.S. player names, approved tournaments, and the 2026 date scope.

5. Analytical Dataset

The central output is a player-centric table in which one row represents one American player's experience in one match. This design makes wins and losses directly comparable and removes the need for downstream analysis to switch repeatedly between winner and loser columns.

Category	Representative fields
Match context	Date, tournament, event type, surface, indoor/outdoor status, round, best-of format, duration, score, and source file.
Player and opponent	IDs, names, countries, seeds, entry types, match-time rankings, and ranking difference.
Outcome	Win/loss, upset win, upset loss, actual result, and performance versus ranking expectation.
Serve performance	Aces, double faults, serve points, first serves in, first-serve win rate, second-serve win rate, ace rate, and double-fault rate.
Break points	Break points faced, saved, save rate, opportunities created, converted, and conversion rate.
Clutch performance	Sets won, straight-set wins, deciding sets, five-set matches, comeback wins, tiebreaks played, and tiebreaks won.

6. Score and Clutch-Time Analysis

The score parser reads set scores from the winner's perspective and extracts completed sets and tiebreak markers. It then converts those results into the selected American player's perspective. This supports analysis of:

- Straight-set wins and losses
- Deciding-set appearances and wins
- Five-set matches and wins at Grand Slams
- Tiebreaks played, won, and win percentage
- Comeback wins after losing the first set
- Retirements, abandoned matches, walkovers, and defaults

Unparseable scores are flagged rather than converted to misleading zeros. Break-point percentages also remain missing when no relevant opportunities occurred.

7. Ranking-Based Expectations

The notebook provides a transparent benchmark for comparing actual results with match-time rankings. The estimated probability of an American win is calculated as opponent rank divided by the sum of the player and opponent rankings. Because a lower numerical ranking is better, this produces a higher expected probability for the better-ranked player.

Measure	Definition
Expected win	Estimated probability of at least 65%.
Toss-up	Estimated probability between 35% and 65%.
Expected loss	Estimated probability of 35% or less.
Performance vs. expectation	Actual result (1 for a win, 0 for a loss) minus the ranking-derived probability.
Interpretation	A positive value indicates overperformance; a negative value indicates underperformance.

This benchmark is intended for interpretation and comparison, not as a calibrated betting or forecasting model. A future phase could replace it with Elo ratings or a trained probability model.

8. Reporting and Visualizations

The notebook contains a player summary table, a configurable match-by-match drill-down, and eight interactive Altair views:

- Win percentage by American player

- Win percentage by surface
- First-serve versus second-serve effectiveness
- Wins over higher-ranked opponents
- Tournament-level match wins
- Cumulative win percentage over time
- Deciding-set and tiebreak performance
- Actual versus ranking-expected win percentage

9. Manual Refresh and Versioning Process

The current notebook is designed for an on-demand refresh rather than an automatic daily schedule. After the source has posted the latest results, the user can open the master notebook, confirm that REFRESH_DATA is True, restart the kernel, and run all cells. The notebook redownloads the two source files, rebuilds the dataset, and updates every output.

- Retain one unchanged master notebook as the reusable template.
- After a successful run, save a dated notebook copy using YYYY-MM-DD in the filename.
- Optionally save a dated processed CSV snapshot to preserve what the source reported that day.
- Review the download messages to confirm that new files were retrieved rather than cached files reused.
- Keep the undated processed CSV as the latest version and dated files as the historical archive.

10. Data Quality and Assumptions

- Tournament filtering uses both event level and an approved event-name list to prevent unrelated ATP events from entering the analysis.
- Live data can overlap the completed-season file; stable match keys and source priority are used to remove duplicates.
- Rankings are the ATP rankings recorded at match time.
- Missing serve statistics are preserved as missing values rather than filled with zero.
- An all-American match correctly appears twice in the player-centric table, but the underlying match remains unique.
- The ranking expectation formula is transparent and easy to explain but is not formally calibrated.
- Source data may be corrected after a match; dated snapshots record the information available at the time of each run.

11. Success Criteria

- Every qualifying 2026 match involving a U.S. player is represented after each refresh.
- No duplicate player-match records remain after processing.
- All retained matches belong to approved Grand Slam or Masters 1000 events.
- Derived serve, break-point, score, and expectation metrics are documented and reproducible.
- Interactive charts update automatically when the underlying dataset changes.
- Dated notebook and CSV snapshots provide an auditable record of the analysis over time.

12. Deliverables

- Master Jupyter Notebook: USA Mens Tennis 2026 Analysis.ipynb
- Cached raw completed-season and ongoing-tournament CSV files
- Processed player-centric CSV: usa_men_big_events_2026.csv
- Player summary and match-level drill-down tables
- Eight interactive Altair visualizations
- Optional dated notebook and CSV snapshots for each refresh date

Conclusion

This project combines sports-domain analysis with a complete analytics workflow: automated ingestion, validation, reshaping, feature engineering, interactive visualization, and repeatable reporting. Its player-centric design makes the data easy to analyze, while the clutch-time and ranking-expectation extensions add meaningful performance context beyond basic win-loss records. The notebook is usable as an ongoing 2026 tracking tool and provides a strong foundation for future Elo modeling or a Streamlit dashboard.